

Mathematical Modeling and Numerical Simulation Supplement

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1 Q1 (final: v0)

Abstract— A one-state numerical model with an approved measured rate.

1.1 Question and Scope

Question. Does x decay under the approved positive k ?

A local one-state approximation.

1.2 Model Assumptions and Applicability

- A-001: k remains constant over the modeled interval. If violated: The rate law must be revised.

1.3 Symbols and Units

ID	Symbol	Meaning	Unit
S-001	x	state	1
S-002	k	rate	s^{-1}

1.4 Mechanistic Equations

$$\frac{dx}{dt} = -kx \quad (1)$$

Q1-EQ-001 where S-001: state; S-002: rate.

1.5 Initial Conditions, Boundary Conditions, and Constraints

Initial Conditions

- $x(0)=1$

Boundary Conditions

- Initial-value condition.

Objective and Constraints

- Compute a bounded finite trajectory.

1.6 Algorithm

Algorithm

Input

- k
- x_0

Output

- $x(t)$

Steps

- Use the fixed ODE adapter.

1.7 Parameters, Scenarios, and Numerical Settings

Parameters

- $k=2 \text{ s}^{-1}$ from the approved parameter set.

Scenarios

- baseline

Solver family. ODE_IVP.

1.8 Parameter Provenance and Applicability

Approved parameter-set identity. a81f3c942c690a736b9460bbc661e38ba2f0390a04493867c6164f9d2fd03216.

- $k(k) = 2.0 \text{ s}^{-1}$; role MATERIAL_PROPERTY; status APPROVED_USER_INPUT; source: Controlled measurement note; document UPD-001; locator: TXT; applicability: required_condition_1=temperature_K; temperature_K=300.0. uncertainty: not reported.

1.9 Numerical Validation and Model-Internal Results

Validation Plan

- Confirm solver convergence.
- Execution sim-1202c9511cb7401496961ae2a7af9187; NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL; relation REFUTED_WITHIN_MODEL; summary: The evidence-bound numerical trajectory decays under the approved parameter set.

1.10 Hypothesis Iteration Lineage

- v0: REFUTED_WITHIN_MODEL (The evidence-bound numerical trajectory decays under the approved parameter set.)

1.11 Limitations, Non-Empirical Disclosure, and References

All values in this section are model-internal numerical simulations (NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL), not empirical observations.

Limitations

- The result is not empirical.

References *Not specified.*